

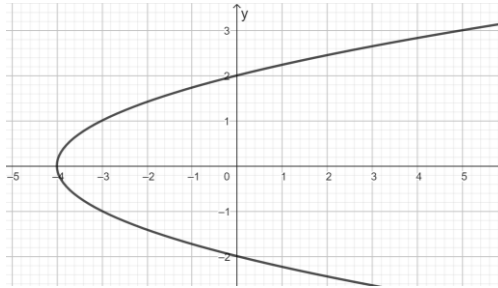
Respuestas del TP 4

1. Parte I:

a) $t \in \mathbb{R}$

b) Dominio: $[-4, \infty)$

c) Gráfica



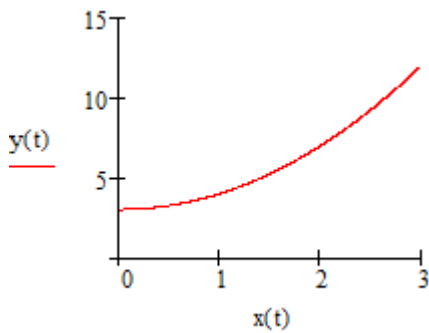
d) $x = y^2 - 4$

Parte II:

a) $t \in [1, \infty)$

b) Dominio: $[0, \infty)$

c) Gráfica



d) $y(x) = 3 + x^2$

2.

A) $y(x) = 3\left(\frac{x-1}{5}\right)^2 - 2\frac{x-1}{5}; x \in \mathbb{R}$

B) $\left(\frac{y+2}{2}\right)^2 + \left(\frac{x-3}{4}\right)^2 = 1, x \in [-1, 7]$

C) $y(x) = \frac{1}{x}, x \in (0, \infty)$

3.

A) $\begin{cases} x(t) = 4 \sin t \\ y(t) = 4 \cos t \end{cases}$

B) $\begin{cases} x(t) = \frac{t}{6} \\ y(t) = \frac{t^2}{12} + 5 \end{cases}$

4.

$$\begin{cases} x(t) = a \cos(\pi - t) \\ y(t) = b \sin(\pi - t) \end{cases}$$

$$\begin{cases} x(t) = a \cos(\pi + t) \\ y(t) = b \sin(\pi + t) \end{cases}$$

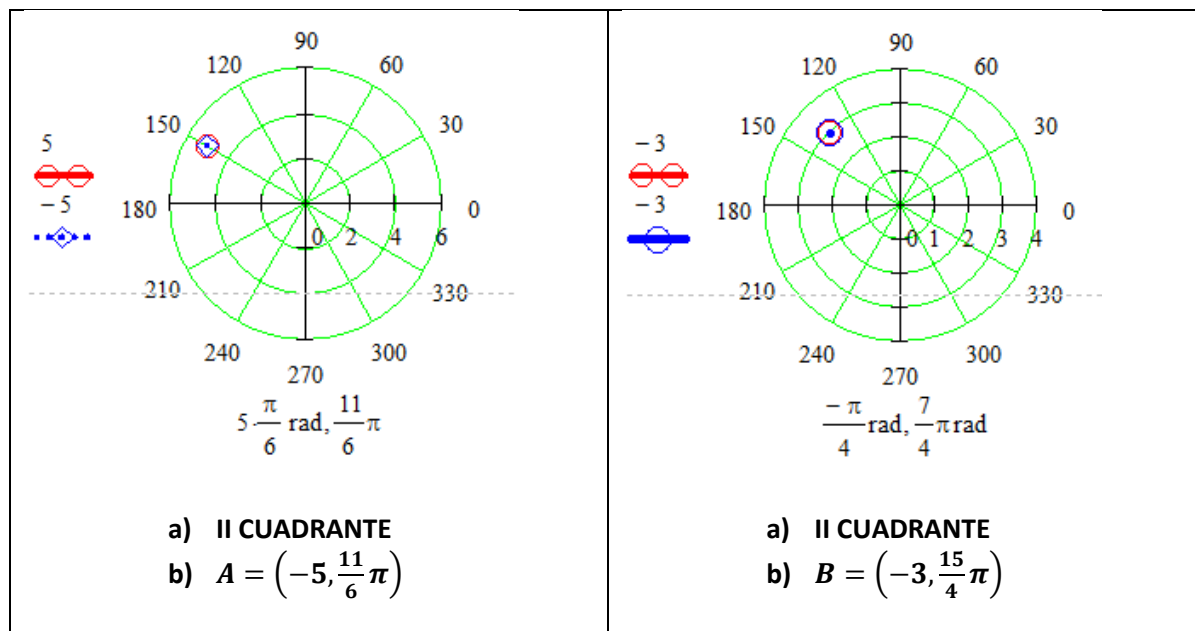
5.

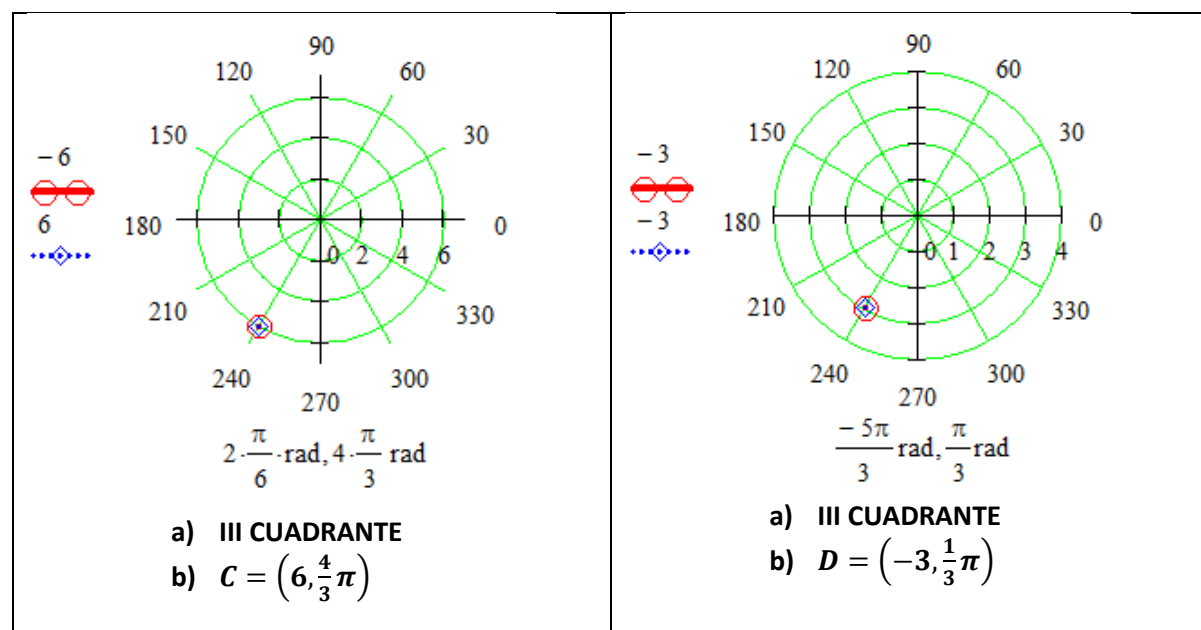
a) $\frac{dx}{dt} = -5$; $\frac{dy}{dt} = 3t^2 + 3$; $\frac{dy}{dx} = -\frac{3}{5}t^2 - \frac{3}{5}$; $\frac{d^2y}{dx^2} = \frac{6}{25}t$

b) $\frac{dx}{dt} = 4 \cos t$; $\frac{dy}{dt} = -2 \sin t$; $\frac{dy}{dx} = -\frac{1}{2} \tan t$; $\frac{d^2y}{dx^2} = -\frac{1}{8} \frac{(1+\tan^2 t)}{\cos t} = -\frac{1}{8} \sec^3 t$

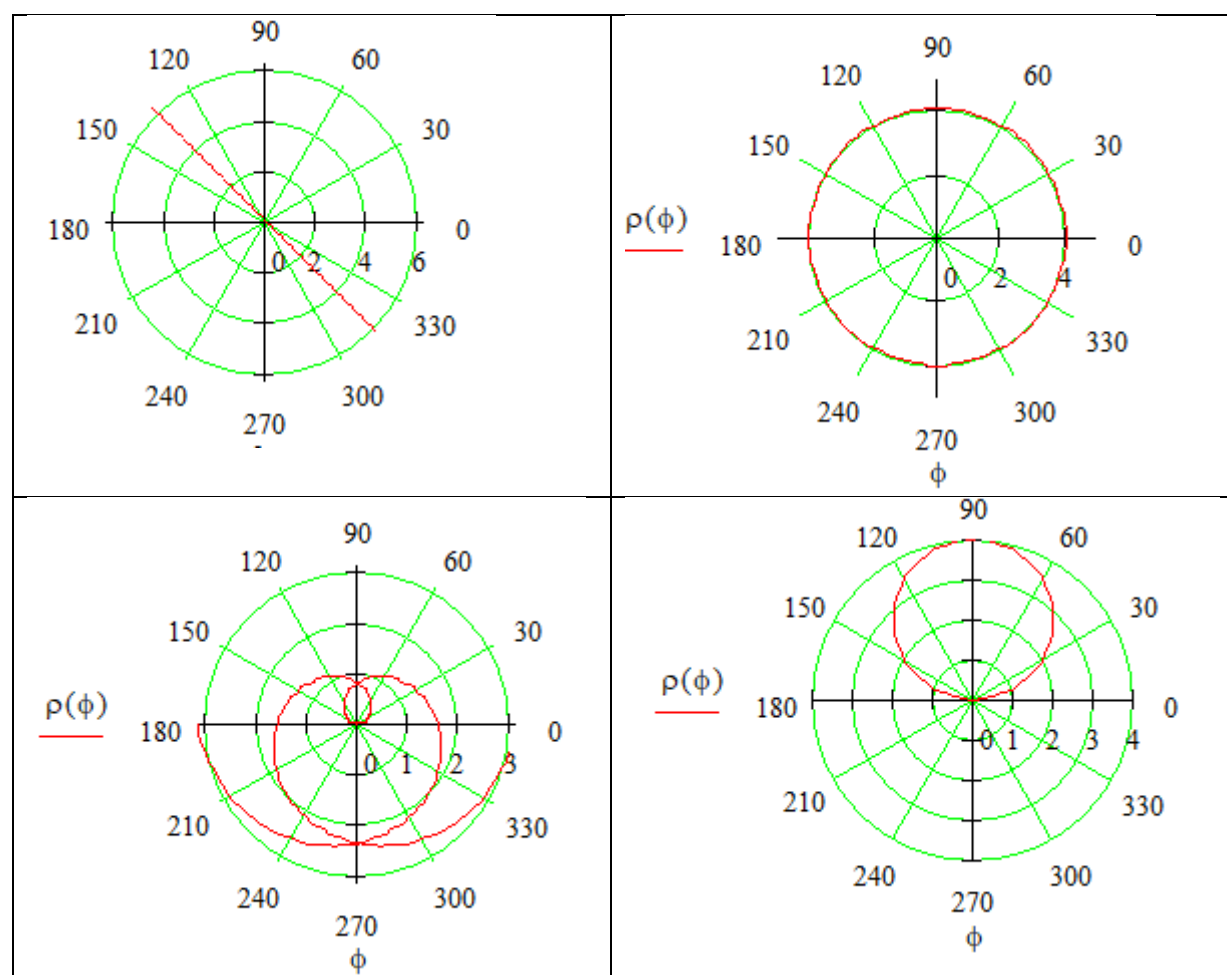
6. $P(1,1)$

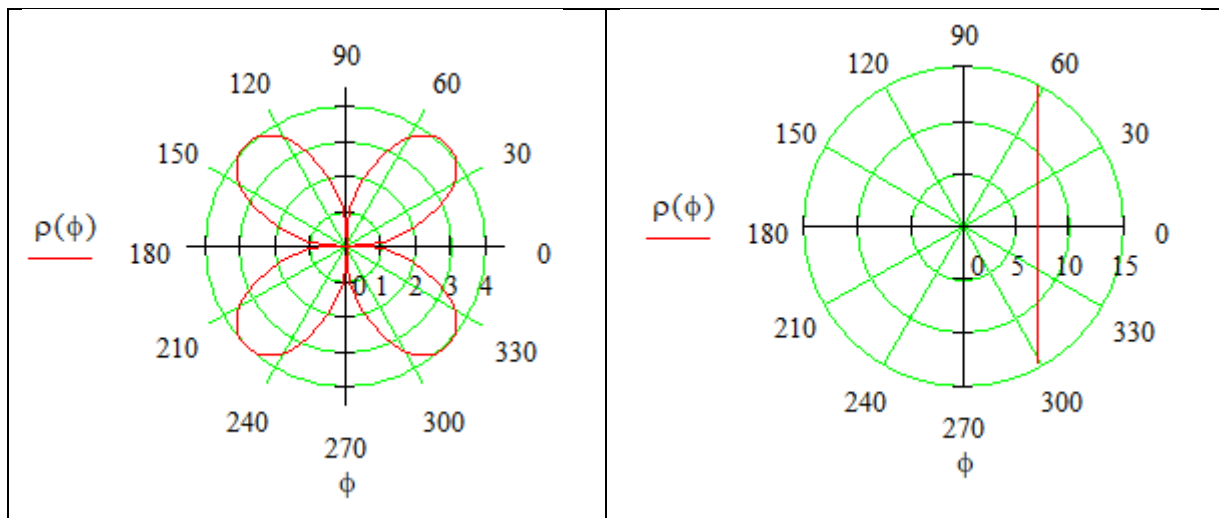
7.





8.





9.

a) Opción **C**

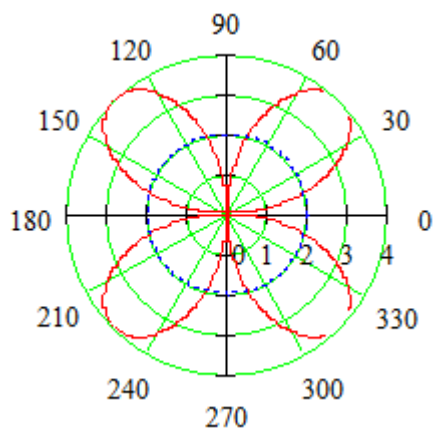
b) Opción **D**

10.

Intersección de curvas

a) En todos los casos $\rho = 2$

$$\theta_1 = \frac{\pi}{12} ; \theta_2 = \frac{5}{12}\pi ; \theta_3 = \frac{7}{12}\pi ; \theta_4 = \frac{11}{12}\pi ; \theta_5 = \frac{13}{12}\pi ; \theta_6 = \frac{17}{12}\pi ; \theta_7 = \frac{19}{12}\pi ; \theta_8 = \frac{23}{12}\pi$$



b) Polo ; $P = \left(\frac{\sqrt{3}}{2}; \frac{1}{6}\pi\right)$

11.

a)

$$\frac{d\rho}{d\theta} = 2 \cos 2\theta \quad ; \quad \frac{dy}{dx} = \frac{2 \cos 2\theta \sin \theta + \sin 2\theta \cos \theta}{2 \cos 2\theta \cos \theta - \sin 2\theta \sin \theta}$$

b)

$$\left. \frac{dy}{dx} \right|_{\frac{\pi}{6}} = -1$$

12.

$$P1\left(1, \frac{1}{2}\pi\right)$$

$$P2\left(3, \frac{3}{2}\pi\right)$$